# Suhwan Choi

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## Summary

I am an undergraduate student majoring in Physics and Computer Science at Seoul National University. My main area of interest is approximating and imitating human behavior and intelligence in multmodal modalities, utilizing end-to-end architectures and scalable training suites.

## Experience

**Principal Researcher** *Feb 2024 – Present*
*Maum.ai*

- Founded autonomous robotics research division as the first researcher, leading strategic decisions and team expansion to 10 researchers.
- Contributed as first author to majority of research projects in robotic navigation and embodied ai.
- Led CORE: Slurm-based DGX Cluster construction project (96 H100 GPUs, 12 nodes). [Blog]
- Implemented company-wide Notion workspace enhancing productivity, streamlining workflows. [Template]

**Machine Learning Engineer Intern** *July 2023 – Jan 2024*
*Hyperconnect*

- Worked on diffusion-based personalized profile image generation for real-world applications. The output images were conditioned not only on prompts but also on the face in the input image.

## Publications

**D2E: Scaling Vision-Action Pretraining on Desktop Data for Transfer to Embodied AI** [Website] 1st author

**Suhwan Choi***, Jaeyoon Jung*, Haebin Seong*, Minchan Kim, Minyeong Kim, Yongjun Cho, Yoonshik Kim, Yubeen Park, Youngjae Yu†, Yunsung Lee†

Under Review

**CANVAS: Commonsense-Aware Navigation System for Intuitive Human-Robot Interaction** [Website] 1st author

**Suhwan Choi***, Yongjun Cho*, Minchan Kim*, Jaeyoon Jung*, Myunchul Joe, Yubeen Park, Minseo Kim, Sungwoong Kim, Sungjae Lee, Hwiseong Park, Jiwan Chung, Youngjae Yu†

ICRA 2025
Workshop on Open-World Agents at NeurIPS 2024 **(Outstanding paper Awards, 3%)**

**Revisiting Residual Connections: Orthogonal Updates for Stable and Efficient Deep Networks** 4th author

Giyeong Oh, Woohyun Cho, Siyeol Kim, **Suhwan Choi**, Youngjae Yu†

NeurIPS 2025

**ESREAL: Exploiting Semantic Reconstruction to Mitigate Hallucinations in Vision-Language Models** 2nd author

Minchan Kim*, Minyeong Kim*, Junik Bae*, **Suhwan Choi**, Sungkyung Kim, Buru Chang†

ECCV 2024

## Education

**Seoul National University** *2021 – Present*
*B.S in Physics & Computer Science and Engineering*

## Awards

**QHack Coding Challenge [link]** *2023 and 2024*

- Ranked **4th/793** teams in 2023, Ranked **3rd/618** teams in 2024.
- This is a contest implementing quantum algorithms, quantum machine learning, quantum chemistry, and brain-

teasing puzzles.

**2023 Quantum Hackathon** [link] *2023*

- **1st place, Minister of Science and ICT Award**
- Topic: Utilizing symmetry to solve variational quantum algorithm (quantum machine learning) efficiently.

**NAVER CLOVA AI RUSH 2022** [link] *July 2022 – Sept 2022*

- **3rd place** on Landmark Detection (3,000,000 KRW)
- **2nd place** on Shopping User Embedding Extraction, Classification (7,000,000 KRW)

**Google Codejam 2022** *2022*

- Round 3, 546th (awarded T-Shirt).

## Activities

**Open Source Projects**

- **open-world-agents**: Core contributor and maintainer with 180+ merged PRs. Built comprehensive multimodal desktop agent framework including optimized data collection tool (ocap), standardized efficient data format (OWAMcap), dataset visualizer, multimedia data management/processing pipelines, agent training and also Python packaging, CI/CD infrastructure.
- **open-world-agents/MediaRef**: Pydantic media reference for images and video frames (with timestamp support) from data URIs, HTTP URLs, file URIs, and local paths. Features lazy loading and optimized batch video decoding.
- **MilkClouds/vla0-trl**: Unofficial reimplementation of VLA-0 using TRL's SFTTrainer. While common VLA codebases are over 10,000 lines, vla0-trl contains only 1,200 lines total. Gets 92% on LIBERO by just fine-tuning Qwen2.5-VL to predict actions as text, without any custom architecture.
- **MilkClouds/smon**: Real-time Slurm cluster monitoring tool with interactive TUI with Textual. Visualizes GPU/CPU/memory allocation across nodes with job-level drill-down.
- **MilkClouds/lazyregistry**: A lightweight Python library for lazy-loading registries with namespace support and type safety

**Open Source Contributions**

- **linkedin/Liger-Kernel** (★6k): Added InternVL3, SmolVLM2 support (**#878**, **#919**).
- **huggingface** (★155k+): Fixed multimodal processor bugs (**transformers#41864**, **#41865**, **#41871**), added ImageProcessor API (**#39603**), PeftModel warning (**#32085**), SDXL batch fix (**diffusers#5807**).
- **open-mmlab** (★40k+): Fixed distributed training deadlocks (**mmdetection#9476**, **mmsegmentation#2398**)

**Technical Blogging**

- Writing tech blog milkclouds.work covering AI/ML research, robotics, industry analysis, and practical engineering insights

**Deepest: SNU Deep Learning Society** [Website] *Sept 2022 – Present*

- Participated in research projects: neural ODE study, RL-based game solving, parallel computing, causal inference, voice generation

**Software Maestro, Trainee** *2021*

- Studied 3D vision and implemented AR game for iPad using AR Foundation of Unity
- Zombie is evading sofa because iPad realized furnitures, and occlusion is also working. [link]

## Skills

**Programming Languages:** Highly proficient in Python / Experience with: C++, Java, HTML/CSS/JS, Golang, Rust, ...

**ML/AI:** PyTorch, HuggingFace (Transformers, Diffusers, LeRobot, ...), open-mmlab libraries, ROS

**Infrastructure & DevOps:** Linux, Slurm, Docker, Ansible, Ceph, GitHub Actions CI/CD, HPC infrastructure

**Languages:** Korean (Native), English